SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

   Assumptions:

- The 4-gallon and 3-gallon jugs can be completely filled from the pump.
- Water can be poured from one jug to the other until one jug becomes full or the other becomes empty.
- Water can be emptied onto the ground whenever required.
- No other measuring device is available.

Solution:

The Water Jug problem is a **state-space search problem** in Artificial Intelligence. Each state represents the amount of water present in both jugs. The objective is to reach a goal state where the **4-gallon jug contains exactly 2 gallons**.

Step	Action Performed	4-Gallon Jug	3-Gallon Jug
1	**Fill the 3-gallon jug from the pump.**	**0**	**3**
2	**Pour the water from the 3-gallon jug into the 4-gallon jug.**	**3**	**0**
3	**Fill the 3-gallon jug again.**	**3**	**3**
4	**Pour water into the 4-gallon jug until it becomes full. Only 1 gallon is required, so 2 gallons remain in the 3-gallon jug.**	**4**	**2**

Step	Action Performed	4-Gallon Jug	3-Gallon Jug
5	**Empty the 4-gallon jug onto the ground.**	**0**	**2**
6	**Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug.**	**2**	**0**

State Diagram

(0,0)
│ Fill 3-gallon
▼
(0,3)
│ Pour to 4-gallon
▼
(3,0)
│ Fill 3-gallon
▼
(3,3)
│ Pour to 4-gallon
▼
(4,2)
│ Empty 4-gallon
▼
(0,2)
│ Pour 2 gallons into 4-gallon
▼
(2,0) ✓ Goal achieved

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
    i. What are the percept's for this agent?
    ii. Characterize the operating environment.
    iii. What are the actions the agent can take?
    iv. How can one evaluate the performance of the agent?
    v. What sort of agent architecture do you think is most suitable for this agent? Why?

Solution:

i. What are the percepts for this agent?
Percepts are the information received by the rover through its sensors. They include:
- Images of the surroundings.
- Rock, soil, and atmospheric data.
- Terrain information, obstacles, and battery status.

ii. Characterize the operating environment
The Mars Rover works in a partially observable, dynamic, and uncertain environment because it cannot sense everything and conditions may change. It is also a sequential and real-world environment, where every action affects the next step.

iii. What are the actions the agent can take?
The rover can:
- Move in different directions and avoid obstacles.
- Capture images and collect rock or soil samples.
- Analyze samples and send the collected data back to Earth.

iv. How can one evaluate the performance of the agent?
The rover's performance is evaluated based on:
- Safe navigation and obstacle avoidance.
- Accuracy of the collected scientific data.
- Efficient use of battery power and successful completion of mission objectives.

v. What sort of agent architecture is most suitable for this agent? Why?

A Goal-Based Agent with Utility-Based features is the most suitable architecture.

A Goal-Based Agent helps the rover achieve its mission, while a Utility-Based Agent selects the best action by considering factors such as safety, battery power, and terrain conditions. This allows the rover to complete its tasks efficiently even in an unpredictable environment.

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

Problem Components

1. Initial State

The chessboard is empty, and no queens are placed.

2. State Space

The state space consists of all possible ways of placing 8 queens on the chessboard. Each state represents a different arrangement of queens.

3. Actions

The action is to place one queen in a safe position in the next column without attacking any previously placed queen.

4. Goal Test

The goal is achieved when:

- All 8 queens are placed on the board.
- No two queens are in the same row, column, or diagonal.

5. Path Cost

Each queen placement has the same cost. Therefore, every step is considered to have a uniform cost.

Search Strategy

Working:

1. Place the first queen in the first column.
2. Move to the next column and place another queen in a safe position.
3. Continue placing queens one by one.
4. If no safe position is available, return to the previous column and try another position.
5. Repeat the process until all 8 queens are placed safely.

State Representation

- **Initial State:** Empty board
- **Intermediate State:** Some queens placed safely.
- **Goal State:** Eight queens placed without attacking each other.

Conclusion

The 8-Queens problem is an important search problem in Artificial Intelligence. It demonstrates how search strategies such as Backtracking and Depth-First Search (DFS) can be used to find a valid solution by exploring different board configurations and eliminating invalid ones. This approach helps in solving complex constraint satisfaction problems efficiently.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Type of Problem-Solving Agent

- The Goal-Based Agent is the most suitable problem-solving agent for this application.
- A Goal-Based Agent works to achieve a specific goal. In this case, the goal is to book a suitable cab and help the customer reach the destination safely and comfortably. The agent considers the customer's pickup location, destination, preferred cab type, and cab availability before making a decision.

Pseudocode:

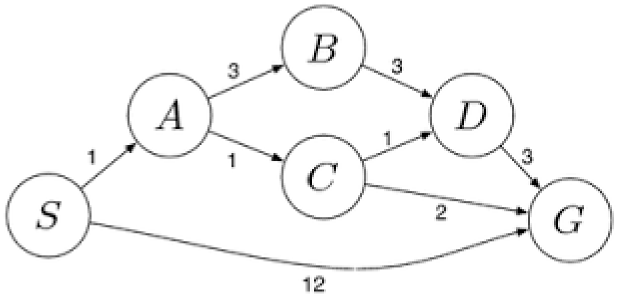
```
Start

Input pickup location
Input destination

Display available cab types
(Mini, Micro, Sedan, Prime, Shared)

Customer selects a cab type

Check cab availability

If cab is available then
    Calculate fare
    Assign nearest driver
    Confirm booking
    Display driver and trip details
Else
    Display "Cab not available"
    Suggest another cab type
End If

End
```

Working :

1. The customer enters the pickup location and destination.
2. The application displays the available cab types.
3. The customer selects the preferred cab.
4. The system checks whether the selected cab is available.
5. If available, the fare is calculated, a nearby driver is assigned, and the booking is confirmed.
6. If the cab is unavailable, the system suggests another available cab.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



Introduction
A logistics company wants to transport packages from the starting warehouse (S) to the destination warehouse (G) with the minimum travel cost. The travel cost includes fuel and time. To find the least-cost path, the Uniform Cost Search (UCS) algorithm is used.

Given Weighted Graph
The edge costs are:

- S → A = **1**
- S → G = **12**
- A → B = **3**
- A → C = **1**
- B → D = **3**
- C → D = **1**
- C → G = **2**
- D → G = **3**

Uniform Cost Search Process

Step 1
Start at **S**.
Open List:

- S (Cost = 0)
  Expand **S**.
  Possible paths:
- S → A = **1**
- S → G = **12**
  Choose **A** (lowest cost).

Step 2
Expand **A**.
New paths:

- S → A → B = **4**
- S → A → C = **2**
  Choose **C** (lowest cost).

Step 3
Expand **C**.
New paths:

- S → A → C → D = **3**
- S → A → C → G = **4**
  Choose **D** (lowest cost).

Step 4
Expand **D**.
New path:

- S → A → C → D → G = **6**

The goal node **G** is already available with a lower cost (**4**), so UCS selects it.

Path Cost Comparison

Path	**Total Cost**
S → G	12
S → A → B → D → G	10
S → A → C → G	4
S → A → C → D → G	6

Least-Cost Path

- **Optimal Path:**
  S → A → C → G

- **Total Cost:**
  1 + 1 + 2 = 4

Using the **Uniform Cost Search (UCS)** algorithm, the least-cost path from the starting warehouse **S** to the destination warehouse **G** is **S → A → C → G** with a total cost of **4**.

Code of Conduct:

I __**K.B.Devarshan/192511426**___ *(Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5